

Rigorous Generalization of a Parameterized Integral Limit

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August 15, 2026

Abstract

In this brief article, we rigorously generalize the limit problem presented in the file `dailyintegral-limit-limit-1x1-2026-08-15.jpg`. We solve a broader class of asymptotic limits involving parameterized definite integrals with changing boundaries and functional exponents. We then apply our main theorem to deduce the precise exact solution to the original problem.

1 Introduction

The problem referenced in `dailyintegral-limit-limit-1x1-2026-08-15.jpg` asks for the value of the following limit:

$$L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \int_n^{2n} x(\log_n(x))^{\frac{1}{n}} dx. \quad (1)$$

We aim to generalize this limit by replacing the monomial powers, the bounding coefficients of integration, and the exponent of the logarithm with arbitrary parameters and asymptotic sequences.

2 Main Theorem

Theorem 1. *Let $a \in \mathbb{R}$, bounds $0 < p < q$, and let $g(n)$ be a real-valued sequence such that $\lim_{n \rightarrow \infty} \frac{g(n)}{\ln n} = c$ for some constant $c \in \mathbb{R}$. Then,*

$$\lim_{n \rightarrow \infty} \frac{1}{n^{a+1}} \int_{pn}^{qn} x^a (\log_n(x))^{g(n)} dx = \int_p^q y^{a+c} dy. \quad (2)$$

This evaluates to $\frac{q^{a+c+1} - p^{a+c+1}}{a+c+1}$ if $a+c+1 \neq 0$, and $\ln\left(\frac{q}{p}\right)$ if $a+c+1 = 0$.

Proof. Consider the sequence of integrals inside the limit:

$$I_n = \frac{1}{n^{a+1}} \int_{pn}^{qn} x^a (\log_n(x))^{g(n)} dx. \quad (3)$$

We apply the substitution $x = ny$. Consequently, $dx = n dy$, and the integration bounds map from $[pn, qn]$ to $[p, q]$. Under this substitution, the integral transforms to:

$$\begin{aligned} I_n &= \frac{1}{n^{a+1}} \int_p^q (ny)^a (\log_n(ny))^{g(n)} n dy \\ &= \int_p^q y^a \left(\frac{\ln(ny)}{\ln n} \right)^{g(n)} dy \\ &= \int_p^q y^a \left(1 + \frac{\ln y}{\ln n} \right)^{g(n)} dy. \end{aligned} \quad (4)$$

We now analyze the pointwise limit of the integrand as $n \rightarrow \infty$. Define $f_n(y) = y^a \left(1 + \frac{\ln y}{\ln n} \right)^{g(n)}$. We express the term dependent on n via the exponential function:

$$\left(1 + \frac{\ln y}{\ln n} \right)^{g(n)} = \exp \left(g(n) \ln \left(1 + \frac{\ln y}{\ln n} \right) \right). \quad (5)$$

For a fixed $y \in [p, q]$, the term $\frac{\ln y}{\ln n}$ approaches 0. Using the Taylor series expansion $\ln(1+u) = u + \mathcal{O}(u^2)$ as $u \rightarrow 0$, we find:

$$\begin{aligned} \lim_{n \rightarrow \infty} g(n) \ln \left(1 + \frac{\ln y}{\ln n} \right) &= \lim_{n \rightarrow \infty} g(n) \left(\frac{\ln y}{\ln n} + \mathcal{O} \left(\frac{1}{\ln^2 n} \right) \right) \\ &= \left(\lim_{n \rightarrow \infty} \frac{g(n)}{\ln n} \right) \ln y. \end{aligned} \quad (6)$$

Given the hypothesis that $\lim_{n \rightarrow \infty} \frac{g(n)}{\ln n} = c$, we obtain:

$$\lim_{n \rightarrow \infty} g(n) \ln \left(1 + \frac{\ln y}{\ln n} \right) = c \ln y. \quad (7)$$

Thus, the pointwise limit of the integrand is:

$$\lim_{n \rightarrow \infty} f_n(y) = y^a \exp(c \ln y) = y^a y^c = y^{a+c}. \quad (8)$$

Because the interval $[p, q]$ is compact and $p > 0$, the sequence of functions $f_n(y)$ is uniformly bounded for sufficiently large n , and it converges uniformly to y^{a+c} . By the Dominated Convergence Theorem, we may interchange the limit and the integral:

$$\lim_{n \rightarrow \infty} I_n = \int_p^q \lim_{n \rightarrow \infty} f_n(y) dy = \int_p^q y^{a+c} dy. \quad (9)$$

This completes the proof of the theorem. □

3 Application to the Original Problem

Corollary 1. *The exact value of the limit shown in the referenced image is $\frac{3}{2}$.*

Proof. Recall the specific limit from `dailyintegral-limit-limit-1x1-2026-08-15.jpg`:

$$L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \int_n^{2n} x (\log_n(x))^{\frac{1}{n}} dx. \quad (10)$$

By matching this to the form established in our Main Theorem, we identify the following parameters:

- Integration bounds: $p = 1, q = 2$
- Power of x : $a = 1$
- Exponent function: $g(n) = \frac{1}{n}$

First, we determine the convergence constant c :

$$c = \lim_{n \rightarrow \infty} \frac{g(n)}{\ln n} = \lim_{n \rightarrow \infty} \frac{1}{n \ln n} = 0. \quad (11)$$

Applying the main theorem with $a = 1$, $p = 1$, $q = 2$, and $c = 0$, the limit evaluates to:

$$\begin{aligned} L &= \int_1^2 y^{1+0} dy \\ &= \int_1^2 y dy \\ &= \left[\frac{y^2}{2} \right]_1^2 \\ &= \frac{2^2}{2} - \frac{1^2}{2} = \frac{4}{2} - \frac{1}{2} = \frac{3}{2}. \end{aligned} \quad (12)$$

□